East West University

Department of Computer Science and Engineering

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Lab Assignment Submission Report-05

ASSIGNMENT TITLE: Introducing Nested Subqueries and Outer Joins in SQL

Course Code: CSE302

Course Title: Database Systems

Section: 02

SUBMITTED BY

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DATE OF SUBMISSION

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Screenshots

1)

--Without using subquery

```
SQL> SELECT customer_name, customer_street, customer_city
  2  FROM customer NATURAL JOIN
  3  depositor NATURAL JOIN
  4  account NATURAL JOIN branch
  5  WHERE branch_city = customer_city;

CUSTOMER_NAME   CUSTOMER_STR CUSTOMER_CITY
--------------- ------------ ---------------
Smith           Main         Rye
Majeris         First        Rye
```

Figure:01

--With using subquery

```
SQL> SELECT customer_name, customer_street, customer_city
  2  FROM(
  3      SELECT customer_name, customer_street, customer_city, branch_city
  4      FROM customer NATURAL JOIN
  5      depositor NATURAL JOIN
  6      account NATURAL JOIN branch
  7      WHERE customer_city = branch_city
  8  );

CUSTOMER_NAME   CUSTOMER_STR CUSTOMER_CITY
--------------- ------------ ---------------
Smith           Main         Rye
Majeris         First        Rye
```

Figure:02

2)

--Without using subquery

```
SQL> SELECT customer_name, customer_street, customer_city
  2  FROM customer NATURAL JOIN
  3  borrower NATURAL JOIN
  4  loan NATURAL JOIN branch
  5  WHERE branch_city = customer_city;

CUSTOMER_NAME   CUSTOMER_STR CUSTOMER_CITY
--------------- ------------ ---------------
Smith           Main         Rye
McBride         Safety       Rye
```

Figure:03

--With using subquery

```
SQL> SELECT customer_name, customer_street, customer_city
  2  FROM customer NATURAL JOIN borrower
  3  WHERE (loan_number, customer_city) IN
  4  (
  5      SELECT loan_number, branch_city
  6      FROM loan NATURAL JOIN branch
  7  );

CUSTOMER_NAME   CUSTOMER_STR CUSTOMER_CITY
--------------- ------------ ---------------
McBride         Safety       Rye
Smith           Main         Rye
```

Figure:04

3)

--With using HAVING clause

```
SQL> SELECT branch_city, avg(balance) as avg_balance
  2  FROM branch NATURAL JOIN account
  3  GROUP BY branch_city
  4  HAVING sum(balance) >= 1000;

BRANCH_CITY     AVG_BALANCE
--------------- -----------
Horseneck             587.5
Brooklyn                625
Rye                   737.5
```

Figure:05

--Without using HAVING clause

```
SQL> SELECT branch_city, avg_balance
  2  FROM(
  3      SELECT branch_city, avg(balance) as avg_balance, sum(balance) as Tot_balance
  4      FROM branch NATURAL JOIN account
  5      GROUP BY branch_city
  6      )
  7  WHERE Tot_balance >= 1000;

BRANCH_CITY     AVG_BALANCE
--------------- ----------
Horseneck            587.5
Brooklyn               625
Rye                  737.5
```

Figure:06

4)

--With using HAVING clause

```
SQL> SELECT branch_city, avg(amount) as avg_amount
  2  FROM branch NATURAL JOIN loan
  3  GROUP BY branch_city
  4  HAVING avg(amount) >= 1500;

BRANCH_CITY     AVG_AMOUNT
--------------- ----------
Palo Alto             2000
Rye                   4035
```

Figure:07

--Without using HAVING clause

```
SQL> SELECT branch_city, avg_amount
  2  FROM(
  3      SELECT branch_city, avg(amount) as avg_amount
  4      FROM branch NATURAL JOIN loan
  5      GROUP BY branch_city
  6      )
  7  WHERE avg_amount >= 1500;

BRANCH_CITY     AVG_AMOUNT
--------------- ----------
Palo Alto             2000
Rye                   4035
```

Figure:08

5)

--With using ALL keyword

```
SQL> SELECT customer_name, customer_street, customer_city
  2  FROM  customer NATURAL JOIN depositor NATURAL JOIN account
  3  WHERE balance >= ALL (
  4      SELECT balance
  5      FROM account
  6  );

CUSTOMER_NAME   CUSTOMER_STR CUSTOMER_CITY
--------------- ------------ ---------------
Johnson         Alma         Palo Alto
```

Figure:09

--Without using ALL keyword

```
SQL> SELECT customer_name, customer_street, customer_city
  2  FROM  customer NATURAL JOIN depositor NATURAL JOIN account
  3  WHERE balance >= (SELECT max(balance)
  4                    FROM account);

CUSTOMER_NAME   CUSTOMER_STR CUSTOMER_CITY
--------------- ------------ ---------------
Johnson         Alma         Palo Alto
```

Figure:10

6)

--With using ALL keyword

```
SQL> SELECT customer_name, customer_street, customer_city
  2  FROM  customer NATURAL JOIN borrower NATURAL JOIN loan
  3  WHERE amount <= ALL (SELECT amount
  4                       FROM loan);

CUSTOMER_NAME   CUSTOMER_STR CUSTOMER_CITY
--------------- ------------ ---------------
Curry           North        Rye
```

Figure:11

--Without using ALL keyword

```
SQL> SELECT customer_name, customer_street, customer_city
  2  FROM  customer NATURAL JOIN borrower NATURAL JOIN loan
  3  WHERE amount <= (SELECT min(amount)
  4                   FROM loan);

CUSTOMER_NAME   CUSTOMER_STR CUSTOMER_CITY
--------------- ------------ ---------------
Curry           North        Rye
```

Figure:12

7)

--With using IN keyword

```
SQL> SELECT DISTINCT branch_name, branch_city
  2  FROM branch
  3  WHERE branch_name IN(SELECT branch_name
  4                       FROM account NATURAL JOIN loan);

BRANCH_NAME     BRANCH_CITY
--------------- ---------------
Downtown        Brooklyn
Redwood         Palo Alto
Perryridge      Horseneck
Mianus          Horseneck
Round Hill      Horseneck
North Town      Rye
Central         Rye

7 rows selected.
```

Figure:13

--With using EXISTS keyword

```
SQL>                     SELECT DISTINCT branch_name, branch_city
  2  FROM branch WHERE EXISTS(
  3      SELECT branch_name
  4      FROM(SELECT *
  5           FROM account NATURAL JOIN loan
  6      )temp
  7  WHERE branch.branch_name = temp.branch_name);

BRANCH_NAME     BRANCH_CITY
--------------- ---------------
Downtown        Brooklyn
Redwood         Palo Alto
Perryridge      Horseneck
Mianus          Horseneck
Round Hill      Horseneck
North Town      Rye
Central         Rye

7 rows selected.
```

Figure:14

8)

--With using NOT IN keyword

```
SQL> SELECT DISTINCT customer_name, customer_city
  2  FROM customer NATURAL JOIN depositor
  3  WHERE customer_name NOT IN (SELECT customer_name
  4                              FROM customer NATURAL JOIN borrower);

CUSTOMER_NAME   CUSTOMER_CITY
--------------- ---------------
Lindsay         Pittsfield
Majeris         Rye
Turner          Stamford
Johnson         Palo Alto
```

Figure:15

--With using NOT EXISTS keyword

```
SQL>                  SELECT DISTINCT customer_name, customer_city
  2  FROM (SELECT *
  3      FROM customer NATURAL JOIN depositor
  4      ) temp1
  5  WHERE NOT EXISTS (SELECT customer_name
  6      FROM (SELECT *
  7            FROM customer NATURAL JOIN borrower
  8            ) temp2
  9  WHERE temp1.customer_name = temp2.customer_name);

CUSTOMER_NAME   CUSTOMER_CITY
--------------- ---------------
Lindsay         Pittsfield
Majeris         Rye
Turner          Stamford
Johnson         Palo Alto
```

Figure:16

9)

--With using WITH clause

```
SQL> WITH branch_balances as (
  2       SELECT branch_name, sum(balance) as Tot_balance
  3       FROM branch NATURAL JOIN account
  4       GROUP BY branch_name
  5   )
  6   SELECT branch_name
  7   FROM branch_balances
  8   WHERE Tot_balance > (SELECT avg(Tot_balance)
  9                        FROM branch_balances);

BRANCH_NAME
---------------
Perryridge
Brighton
Central
```

Figure:17

--Without using WITH clause

```
SQL> SELECT branch_name
  2   FROM (
  3       SELECT branch_name, sum(balance) as Tot_balance
  4       FROM branch NATURAL JOIN account
  5       GROUP BY branch_name
  6   )
  7   WHERE Tot_balance > (
  8       SELECT avg(Tot_balance)
  9       FROM (
 10           SELECT branch_name, sum(balance) as Tot_balance
 11           FROM branch NATURAL JOIN account
 12           GROUP BY branch_name
 13           )
 14   );

BRANCH_NAME
---------------
Perryridge
Brighton
Central
```

Figure:18

10)

--With using WITH clause

```
SQL> WITH branch_amounts as (
  2      SELECT branch_name, sum(amount) as Tot_amount
  3      FROM branch NATURAL JOIN loan
  4      GROUP BY branch_name
  5  )
  6  SELECT branch_name
  7  FROM branch_amounts
  8  WHERE Tot_amount < (SELECT avg(Tot_amount)
  9                      FROM branch_amounts);

BRANCH_NAME
---------------
Round Hill
Mianus
Redwood
Central
```

Figure:19

--Without using WITH clause

```
SQL> SELECT branch_name
  2  FROM (
  3      SELECT branch_name, sum(amount) as Tot_amount
  4      FROM branch NATURAL JOIN loan
  5      GROUP BY branch_name
  6  )
  7  WHERE Tot_amount < (SELECT avg(Tot_amount)
  8                      FROM(
  9                       SELECT branch_name, sum(amount) as Tot_amount
 10                       FROM branch NATURAL JOIN loan
 11                       GROUP BY branch_name)
 12  );

BRANCH_NAME
--------------
Round Hill
Mianus
Redwood
Central
```

Figure:20

Learning Outcomes: The following is what we have learned in this lab. We have learned to Subqueries in the WHERE clause namely (IN / NOT IN, SOME / ALL and EXISTS/NOT EXISTS). We have also learned Subqueries in the FROM clause, complex queries using WITH clause and also the SELECT clause (Scalar Subquery). Then finally, we have completed the lab tasks.